Collective Semantic Memory: Merge, Trust, and Staleness for Peer Memory in Multi-Agent Navigation

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Abstract

Problem. Collective memory is often treated as monotone: more shared information should help. In cooperative navigation with scarce semantic targets, this intuition fails. Fast global sharing improves target materialization but causes agents to race for the same targets; slow sharing avoids contention but leaves agents with stale or incomplete memories. **Approach.** We instantiate four memory topologies over the same symbolic-semantic substrate: independent private memories, a shared event bus, a central consensus aggregator, and peer-to-peer broadcast with cadence K. We then introduce a minimal *collective semantic memory* (CSM): peer broadcast plus explicit *merge*, *trust*, and *staleness* rules. Q/R/M/C stage rates are used only as measurement instrumentation, not as the contribution of this paper. **Findings.** On a 35,640-run multi-agent sweep, the fixed-cadence peer family exhibits a bottleneck shift: fast broadcast saturates materialization but collapses completion, while slow broadcast has the opposite profile. Mean time-to-success has an interior optimum at $K{=}8$. On a 3,240-run scarcity benchmark, the minimal CSM strictly dominates five fixed-cadence peer baselines on success rate: CSM reaches 0.617 with 95% CI $[0.610, 0.624]$, above every peer-K upper CI. **Scope.** Experiments are grid-world substrates. The claim is not that this is a complete theory of collective memory; it is a reproducible architecture split showing that merge/trust/staleness are the right explicit control surfaces after cadence alone is exhausted.

1 Introduction

Many multi-agent systems share information either by writing to a single global memory, by sending learned messages, or by periodically exchanging local state. These mechanisms are usually evaluated end-to-end. If success improves, communication is judged useful; if it does not, the reason is opaque. This paper studies a narrower architectural question: when multiple agents maintain semantic memory about places, what should be shared, at what rate, and with what weight?

The answer is not "share everything as soon as possible." In a scarcity setting, where N agents compete for $T < N$ semantic targets, fresh global memory can make all agents select the same target. The memory is then accurate but unhelpful. Conversely, no communication prevents herding but starves agents of useful discoveries made by peers. The design problem is therefore a trade-off between semantic freshness and occupancy contention.

We make three contributions.

1. **Architecture split.** We implement and compare four memory topologies over the same symbolic substrate: independent, shared-bus, central-aggregator, and peer-to-peer broadcast.

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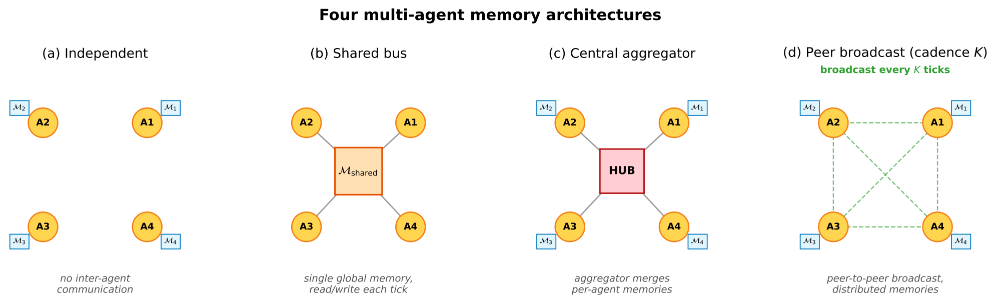

Figure 1: Four memory topologies. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads and writes each tick. **(c)** Central aggregator: private memories plus a hub that merges snapshots into a consensus report. **(d)** Peer-to-peer broadcast: private memories with snapshot exchange every K ticks and per-agent merged views.

2. **Cadence result.** We show that peer broadcast exposes a real coordination parameter K: fast broadcast improves $P(M^\star|R^\star)$ but hurts $P(C^\star|M^\star)$; the efficiency metric has an interior optimum.
3. **Minimal CSM.** We instantiate collective semantic memory with three explicit rules: merge, trust, and staleness. The minimal CSM reaches a new point on the M×C plane and strictly dominates fixed-cadence peer baselines on success rate in the scarcity benchmark.

Relation to the companion Q/R/M/C paper. The companion paper treats Q/R/M/C as a diagnostic framework. Here those stage rates are used as measuring instruments. The main object is different: the design of distributed symbolic memory itself.

2 Memory substrate and task

Each agent observes a grid environment and emits semantic place evidence such as `water_source`, `hazard_edge`, and `open`. A local memory stores place records, tag confidences, visit counts, freshness, and transition statistics. Planner queries retrieve candidate places by matching required, preferred, and penalty tags. A selected candidate is materialized into a concrete cell and executed by a local controller.

The multi-agent task uses a 12×10 grid. Each agent must reach a cell tagged `water_source`. Targets are scarce: $N\in\{3,5,8\}$ agents, $T\leq N$ targets, three layouts, three hazard densities, and 40 seeds for the main cadence sweep. Every run logs Q/R/M/C events per agent:

$$Q^\star \to R^\star \to M^\star \to C^\star,$$

where $M^\star$ means a semantic candidate was locked as an actionable target and $C^\star$ means the target was reached. These logs are not the architectural contribution; they let us see where each topology fails.

3 Four collective-memory topologies

The four topologies share the same local memory and planner. They differ only in where merge happens.

Independent. Each agent owns a private event store and concept graph. There is no communication path.

Shared bus. All observations are appended to a single global event bus. One shared concept graph is rebuilt from the union of all events.

Central aggregator. Each agent owns a private concept graph; a central consensus layer pulls snapshots and computes one trust-weighted report for everyone.

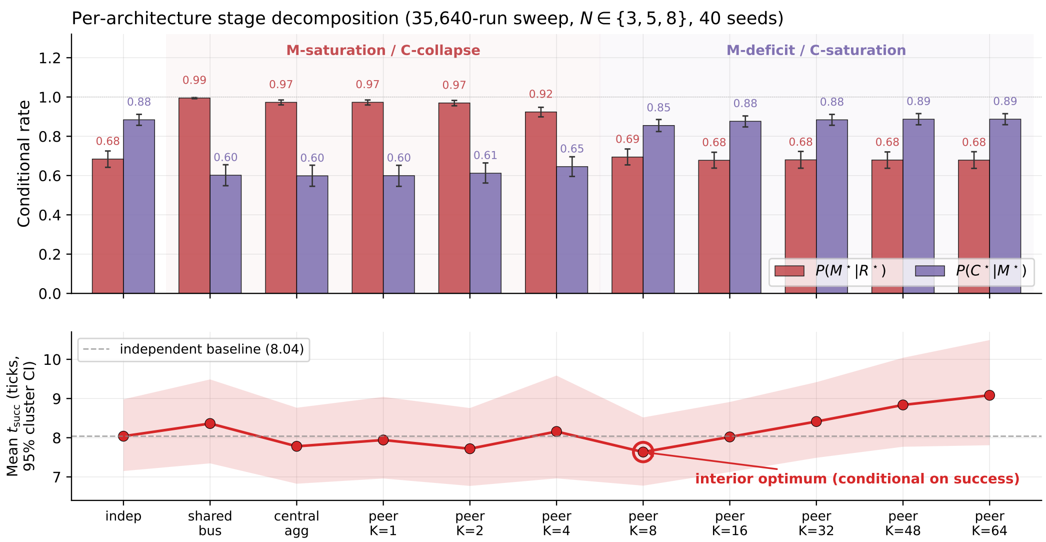

Figure 2: Conditional rates and mean time-to-success from the 35,640-run sweep. Fast sharing improves target materialization but harms completion through contention; slow sharing has the opposite profile. Mean time-to-success has an interior optimum at $K{=}8$.

Peer broadcast. Each agent owns a private concept graph and a private merged view. Every K ticks, agents broadcast snapshots; each agent merges its own memory with the latest snapshot from each peer.

The key distinction is not the clustering primitive. It is the level at which that primitive is instantiated:

$$\begin{aligned} \text{independent} &: N \text{ graphs, } 0 \text{ mergers} \\ \text{shared bus} &: 1 \text{ graph, } 1 \text{ implicit global merger} \\ \text{central} &: N \text{ graphs } + 1 \text{ central merger} \\ \text{peer} &: N \text{ graphs } + N \text{ private mergers.} \end{aligned}$$

4 Cadence exposes the failure mode

The peer topology has one explicit parameter: broadcast cadence K. The main sweep crosses four architectures and eight peer cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$ for 35,640 runs. Figure 2 summarizes the result.

Fast peer broadcast and centralized sharing live in the M-saturation/C-collapse regime: agents can lock targets reliably, but many locks point to targets that peers also pursue. Independent and slow peer broadcast live in the M-deficit/C-saturation regime: fewer agents lock onto good targets, but those that do are less likely to collide with peer occupancy.

Across all peer runs, Spearman of $P(M^\star|R^\star)$ vs K is -0.608 and of $P(C^\star|M^\star)$ vs K is $+0.420$ (both $p < 10^{-4}$). The stronger signature is visible at fixed cadence: the within-cadence Pearson correlation between the M and C links peaks at -0.61 for $K{=}4$ and decays toward noise as K grows. Cadence therefore does not simply tune "amount of communication." It moves the bottleneck between semantic commitment and shared-resource completion.

5 Minimal collective semantic memory

The cadence sweep shows that fixed-rate snapshot exchange exposes only one control surface. A collective semantic memory should expose at least three:

Merge. How peer evidence about the same semantic place is combined.

Trust. How evidence from different peers is weighted when it conflicts or becomes unreliable.

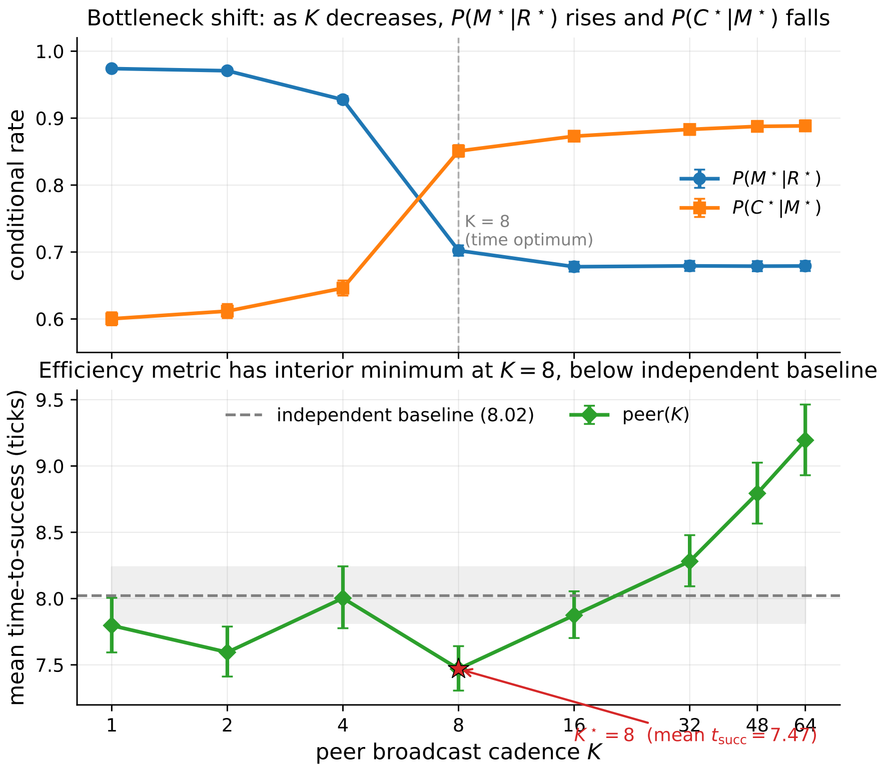

Figure 3: Bottleneck shift across peer cadences. $P(M^\star|R^\star)$ falls and $P(C^\star|M^\star)$ rises as K grows. The practical efficiency metric, mean time-to-success, is minimized at K=8.

Staleness. How old evidence decays before it influences retrieval.

The minimal CSM uses peer broadcast at K=8 and overlays three rules:

Merge. Per-place score equals own evidence at weight 1 plus peer evidence at weight $\text{trust}_i \cdot e^{-\alpha \cdot \text{age}}$, with $\alpha = 0.05$. Inclusion threshold $\tau = 0.30$.

Trust. Per-peer EMA on retrieval-success consistency. Peers whose latest snapshot supports the locally locked target gain trust; others decay toward 0.4. Update rate $\beta = 0.10$.

Staleness. Each broadcast snapshot is timestamped; merge weight decays as $e^{-\alpha \cdot \text{age}}$. Snapshots older than $10K$ ticks are pruned.

The implementation is a drop-in replacement for fixed-cadence peer memory: `experiments/collective_semantic_memory/csm_memory.py`.

5.1 Results

We compare CSM against fixed-cadence peer broadcast at $K \in \{1, 4, 8, 16, 64\}$ on scarcity cells $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, three layouts, three hazard densities, and 20 seeds: 3,240 runs total.

CSM occupies a third regime. Fast broadcasts have excellent materialization but poor completion; slow broadcasts have the reverse. CSM keeps both conditional links above 0.78. The absolute success gain is modest, +1.7 to +4.0 percentage points, but the stage decomposition shows why it is meaningful: the architecture changes the joint M×C operating point rather than merely moving along the fixed-cadence curve.

Why CSM wins: the gate, not the sharing. Provenance annotation of the same runs gives the mechanism. CSM does not share *better* than fixed-cadence peers — it *refuses to act on stale or untrusted foreign evidence*. The inclusion rule of Section 5 implies a closed-form horizon beyond

Table 1: Minimal CSM vs fixed-cadence peer broadcast on the scarcity benchmark. CSM's lower 95% bootstrap CI bound exceeds every peer-K upper CI.

Architecture	$P(M^\star \mid R^\star)$	$P(C^\star \mid M^\star)$	Success rate (95% CI)
peer (K=1)	0.991	0.586	0.577 [0.572, 0.586]
peer (K=4)	0.955	0.623	0.581 [0.572, 0.590]
peer (K=8)	0.730	0.871	0.599 [0.591, 0.607]
peer (K=16)	0.697	0.892	0.597 [0.589, 0.606]
peer (K=64)	0.684	0.900	0.601 [0.594, 0.609]
CSM (minimal)	**0.782**	**0.825**	**0.617 [0.610, 0.624]**

which a peer's evidence can no longer form an intention, $\mathrm{age}_{\max}(\mathrm{trust}_i) = \alpha^{-1}\ln(\mathrm{trust}_i \cdot c/\tau)$, and this horizon is empirically exact: measured intention-formation cutoffs match the formula tick-for-tick across trust levels (23/23.05, 18/18.59, 12/12.84 at $\mathrm{trust} = 1.0/0.8/0.6$), including a registered-prediction hold-out on six never-run cells ($6/6$ exact integer breakpoints) and a discrete trust-flip: $\mathrm{trust} \leq \tau/c$ zeroes foreign-evidence commitments entirely. Fixed-cadence peers have no such gate — their foreign evidence never expires — which is precisely why their completion collapses when sharing is fast. The full provenance-conditioned analysis appears in a companion manuscript; here it serves as the mechanistic reading of Table 1.

6 Place identity from meaning: removing the shared frame

Everything above — and the categorical reading below — assumes a shared spatial basis X: 'the same place' means 'the same (x, y) in a frame the environment hands to every agent'. This section removes that assumption and reports what it was hiding. Each agent now holds a *private* frame (the true grid under a secret translation, and, in the final experiment, a secret $90°$-multiple rotation); broadcast snapshots carry the sender's private coordinates, which are meaningless to the receiver until it decides what the sender's places *mean*.

Mechanism. A *place fingerprint* is the local semantic constellation around a visited cell: which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Two agents match places by *mutually unique* fingerprint correspondence (each side is the other's sole candidate; one-directional uniqueness aliases distant look-alikes into the receiver's neighbourhood, and border-only patterns are excluded as descriptions of the world's shape rather than of a place). The inter-agent frame is the modal translation of ≥ 3 agreeing matched pairs; under rotations, one hypothesis per $90°$ reuses the same aligner and the winner must dominate the runner-up's consensus by $2\times$. Foreign evidence — including places the receiver has never seen — is then transported through the recovered frame.

Results. Registered before execution; full protocol in the companion manuscript.

1. **Exact recovery.** Witness–traveler: the secret offset is recovered exactly in 100% of landmark-world runs, and under full $SE(2)$ frames in $80/80$ rotation$\times$offset$\times$seed combinations, with 100% completion on foreign-committed locks.
2. **Failure asymmetry.** Where identification is impossible — a featureless region, or a $180°$-symmetric layout in which the frame is unrecoverable *in principle* — semantic identity makes *zero* commitments (fails closed), while coordinate identity keeps confidently locking wrong cells (100% wrong locks; fails open).
3. **The full stack.** With N=4 agents in distinct secret frames, alignment develops online (onset in 100% of episodes, mean tick $\approx$15) and recovers 88% of the shared-frame performance gap (success 0.444 vs. 0.456 oracle vs. 0.350 coordinate); under random $SE(2)$ frames every semantic commitment points at real water (8/8) while the coordinate stack produces 992 phantom locks (0 real).
4. **The gate is orthogonal.** Re-running the $\mathrm{age}_{\max}$ analysis above with the CSM inclusion rule ON TOP of semantic identity, under a frame offset and under a $90°$ rotation, reproduces the SAME integer breakpoints (20/14/9; $\mathrm{bp} = 20$ rotated) — the trust$\times$staleness gate and frame recovery are independent layers.

The stronger claim promised in earlier drafts' limitations is therefore discharged constructively: place identity can be carried by meaning alone, it composes with merge/trust/staleness unchanged, and — unlike the coordinate convention it replaces — it knows when it does not know.

The categorical indicator: adjunction round-trip defect. Frame recovery is exactly the operation the categorical reading of §7 needs once the shared basis X is dropped: with private frames each agent carries its own place category, and communication is a pair of recovery functors $F = \textsc{align}(A, B)$ (the sender B's frame expressed in receiver A's coordinates) and $G = \textsc{align}(B, A)$. If the two are genuinely adjoint — the frames are glued — the round trip $\eta_X : X \to G(F(X))$ is the identity, so the *adjunction defect*

$$\mathrm{def} = \|F.\mathrm{offset}{+}G.\mathrm{offset}\|_1 \text{ (translation)}, \qquad r_F{+}r_G \equiv 0 \pmod 4 \text{ and } \|R_{r_G}F.\delta{+}G.\delta\|_1 \ (SE(2)),$$

is zero (Mac Lane, 1998). Measured on the recovery method above (`experiments/warp/alignment_defect.py`, 3 offsets $\times$ 10 seeds per cell): the defect is *exactly* 0 in $30/30$ runs for translation frames and $30/30$ for each of the four $90°$ rotations — the recovered functors are strict adjoints. Where recovery must fail (an agent that has observed only a featureless interior band), F or G is undefined in $30/30$ runs with *zero* spurious gluings ($\mathrm{def} > 0$): no adjoint exists and the aligner refuses to build one. The exact discrete aligner therefore realizes a dichotomy — strict adjoint ($\mathrm{def}{=}0$) or no adjunction — and never a confident wrong gluing; "failing closed" above is precisely *refusing a non-adjoint pair*. This is the quantity the categorical model computes that a re-description does not: a scalar that is zero iff the collective's local frames are coherently glued, and it is this gluing — not a shared coordinate convention — that §7's merge silently presupposed.

7 Categorical BDI reading of collective semantic memory

This section gives a compact theoretical reading of the water-search example in terms of belief, desire, and intention. The goal is not to introduce a new planner, but to make explicit the mathematical structure already used by the collective semantic memory architecture.

We assume a finite set of agents $A = \{1, \ldots, N\}$, a shared spatial basis X, and a shared semantic alphabet Σ. In the present grid-world implementation, X is the common coordinate frame and Σ contains tags such as `water_source`, `water_candidate`, `hazard_edge`, and `open`. This is an explicit modelling assumption. The agents may have different observations, different memories, and different confidence values, but they interpret places in a common coordinate framework and express semantic evidence in a common tag alphabet. Without the first assumption, peer merge requires a place-alignment problem to be solved first — Section 6 solves it constructively (fingerprint consensus recovers the inter-agent frame), and Section 7.1 states what that construction is, categorically. Symbol alignment (a shared Σ) remains assumed.

What parametrises commonality of interpretation. It is worth stating explicitly that the categorical setup localises "how much the agents mean the same thing" in exactly three parameters with three different statuses. *(i) The alphabet* Σ — what counts as the same *predicate* — is a parameter of the category itself (the subscript of $\mathbf{Ctx}_\Sigma$) and is here *assumed*; relaxing it means constructing a translation between $\mathbf{Ctx}_{\Sigma_1}$ and $\mathbf{Ctx}_{\Sigma_2}$, which we leave open. *(ii) The alignment morphisms* of the merge diagram — what counts as the same *place* — were assumed in earlier drafts and are now *constructed* (Sections 6 and 7.1), with their existence an empirically decidable predicate. *(iii) The trust$\times$staleness enrichment* — how much a peer's interpretation *counts* — makes commonality graded rather than binary, with its own closed-form horizon and a discrete trust-flip. The three are ordered by hardness: weights are tunable, place correspondence is solvable from co-observation, and predicate correspondence is genuinely open. The same triple reappears one level up, at the evaluator: scoring retrieval and materialization presupposes a correspondence between memory entities and world referents (see the companion measurement paper's limitations), and grounding *that* correspondence by the mechanisms of Section 6 is future work.

Parameter (i) has, in addition, a measurable *interior*: within a fixed Σ, which tag classes actually carry place correspondence is an empirical property, not a modelling choice. A registered vocabulary ablation in the companion route-transfer paper makes the alphabet itself the treatment and finds the identity machinery *fail-safe in* Σ: impoverishing the vocabulary down to a single content class converts frame recovery into refusal, never into a wrong alignment (zero fail-open transports across 252 cells), and the consensus mass is attributable per class (hazards 0.56, walls 0.44, world borders 0.00). What no mechanism yet does is *select* Σ: here features receive landmark status from design plus the world's cooperation. A rule by which features *earn* that status through participation in successful alignments — the alphabet shaped by coordination rather than assumed — is the sharpest open end of parameter (i), and the natural meeting point with the symbol-alignment programme of §7.4.

At time t, agent i maintains a local semantic context $\mathcal{K}_t^i = (G_t^i, \Sigma, I_t^i)$, where G_t^i is the set of place records stored in the agent's private memory, and

$$I_t^i \subseteq G_t^i \times \Sigma$$

is the thresholded incidence relation saying that place $g \in G_t^i$ carries tag $\sigma \in \Sigma$ with sufficient confidence. The graded implementation uses tag confidences and weighted scores; the crisp relation I_t^i is the thresholded skeleton needed for the theoretical description.

This formal context induces a Galois connection between sets of tags and sets of places:

$$\mathrm{ext}_t^i(B) = \{g \in G_t^i : \forall \sigma \in B,\ (g, \sigma) \in I_t^i\}, \qquad \mathrm{int}_t^i(S) = \{\sigma \in \Sigma : \forall g \in S,\ (g, \sigma) \in I_t^i\}.$$

For a water-search task, the desired semantic predicate is $B_{\mathrm{water}} = \{\texttt{water\_source}\}$, or, in the implemented weighted form, a predicate combining positive evidence for `water_source`, `water_candidate`, and `water_nearby` with penalties for `hazard_edge` or nearby risk. Retrieval succeeds for agent i at time t exactly when

$$\mathrm{ext}_t^i(B_{\mathrm{water}}) \neq \varnothing.$$

Thus the retrieval stage is not a coordinate lookup. It is the test that the queried semantic meaning already has a non-empty extent in the agent's realized memory.

We can now define the BDI components.

Belief. The belief state of agent i is the structured semantic memory induced by $\mathcal{K}_t^i$. Equivalently, it is the concept lattice $L_t^i = \mathrm{Concepts}(\mathcal{K}_t^i)$, whose nodes pair extents of places with intensions of tags. A water belief is not the proposition "there is water at coordinate x." It is the region of L_t^i whose intension contains the water predicate. If this region is empty, the agent does not merely choose badly; the relevant semantic object is absent from its current memory.

Desire. Desire is a state-conditioned semantic predicate over the belief lattice. In the water example, the agent's internal state contains a thirst variable. The active desire can be written as

$$D_t^i = \begin{cases} B_{\mathrm{water}}, & \text{if } \mathrm{thirst}_t^i \geq \theta_{\mathrm{on}}, \\ B_{\mathrm{goal}}, & \text{otherwise.} \end{cases}$$

Thus desire is not an additional metaphysical object. It is the rule that selects which semantic predicate should be queried from the current belief structure.

Intention. Intention is the commitment to one materialized candidate. Once desire has selected a predicate and retrieval has produced a non-empty extent, the planner applies a satisficing choice function

$$\chi_t^i : \mathcal{P}(G_t^i) \to G_t^i \cup \{\perp\}, \qquad g_t^{i,\star} = \chi_t^i\big(\mathrm{ext}_t^i(D_t^i)\big),$$

where $\perp$ denotes failure to materialize a target. In Q/R/M/C terms, desire determines the query $Q^\star$, belief supports retrieval $R^\star$, and intention corresponds to materialization $M^\star$. Completion $C^\star$ is downstream of this semantic structure: it depends on whether the physical controller reaches the locked target and whether the target remains available under multi-agent occupancy.

This gives a compact categorical reading. Let $\mathbf{Ctx}_\Sigma$ be the category whose objects are semantic contexts (G, Σ, I) over the fixed alphabet Σ, and whose morphisms preserve semantic incidence. A local memory trajectory for agent i is a time-indexed diagram

$$\mathcal{K}_0^i \to \mathcal{K}_1^i \to \cdots \to \mathcal{K}_t^i \to \mathcal{K}_{t+1}^i \to \cdots$$

in $\mathbf{Ctx}_\Sigma$. Each arrow is induced by observation and memory update. When new evidence is consolidated, the formal context changes; therefore its concept lattice changes. This is the order-theoretic content of memory growth.

In Q/R/M/C terms, an empty extent $\mathrm{ext}_t^i(D_t^i) = \varnothing$ is a belief/retrieval failure, a non-empty extent with no stable lock is a materialization failure, and a successful lock that cannot be reached is a completion failure — the semantic and the physical parts of the agent's state fail in different, separately measurable ways.

The collective case lifts this by one construction. At broadcast times the receiver i holds a diagram of local memory objects $\mathcal{D}_t^i = (\mathcal{K}_t^i, \mathcal{K}_t^{j_1}, \dots, \mathcal{K}_t^{j_m})$ together with alignment maps, and queries the merged consensus object $\widetilde{\mathcal{K}}_t^i = \mathrm{Merge}_{\mathrm{trust,stale}}(\mathcal{D}_t^i)$, with peer evidence weighted exactly by the rules of Section 5 ($\mathrm{trust}_{j\to i} \cdot e^{-\alpha\, \mathrm{age}_j}$). The same BDI definitions apply to the merged context: intention is now selected from $\mathrm{ext}_{\widetilde{\mathcal{K}}_t^i}(D_t^i)$ rather than from the private extent alone.

This explains why communication cadence produces a bottleneck shift. Faster broadcast increases the freshness of peer evidence. This tends to enlarge or improve the water extent $\mathrm{ext}_{\widetilde{\mathcal{K}}_t^i}(B_{\mathrm{water}})$, so agents can materialize targets more reliably; in Q/R/M/C terms, this raises $P(M^\star \mid R^\star)$. But the same fresh shared belief also synchronizes intentions. Several agents may lock onto the same scarce water source. The memory is semantically accurate, but the physical target is now contested. This lowers $P(C^\star \mid M^\star)$. The bottleneck is therefore physically grounded. It is not a formal artifact of the decomposition: it arises because semantic belief is shared through memory, while completion is constrained by occupancy and scarce resources. Collective semantic memory must therefore regulate not only how much information is shared, but also how it is merged, whose evidence is trusted, and when stale evidence should stop influencing intention formation.

In this sense the "collective" is not a central mind and not a global map but a stabilised family of peer-indexed processes; the next subsection makes that statement precise.

The dual, process-level reading — agents as coalgebras (codata), the collective as a final coalgebra, communication as depth-one corecursion, and the bottleneck shift as a competition between belief- and intention-bisimilarity — is developed in Appendix A. It is a lens, not a mechanism: nothing downstream depends on it, which is why it lives in the appendix.

7.1 Colimit merge and its invariants

The merge object above can be described more precisely. Working in a timestamp- and provenance-enriched version of $\mathbf{Ctx}_\Sigma$, a snapshot merge from agents $1, \dots, N$ is a diagram of local memory objects plus alignment maps induced by the place-identity relation. The consensus object $\widetilde{\mathcal{K}}_t^i$ is the colimit of this diagram, implemented concretely by the union-find merge over aligned concepts. This reading gives three useful constraints.

1. **Order independence.** The merged object is unique up to isomorphism, so the result is not an artefact of processing agent snapshots in a particular order.
2. **Chaining is expected.** If concept A aligns with B and B aligns with C, union-find performs the transitive closure. Over-merging is therefore not a bug in the data structure; it is the cost of the chosen equivalence relation.
3. **Decay is not a morphism.** Staleness is kept as a weight field outside the alignment morphisms. Otherwise age-then-merge and merge-then-age need not commute, breaking the clean interpretation of broadcast snapshots as one diagram.

In earlier drafts this section ended by conceding that the model does not answer which perceptual features should identify two places — the alignment maps were assumed, not constructed. That gap is now closed by Section 6, and the closure gives the colimit reading its missing scientific function:

1. **The alignment morphisms are constructed, not assumed.** Fingerprint consensus IS an algorithm that produces the alignment maps of the diagram — mutually-unique landmark matches determine the frame transform through which every local memory object embeds.
2. **Existence becomes an empirical predicate.** The colimit exists iff the consensus does. In a $180°$-symmetric world the alignment morphism does not exist *in principle* (two incomparable diagrams support equal consensus), and the implementation detects exactly this and refuses to merge — the fail-closed behaviour of Section 6 is the categorical non-existence made operational. This predicate is measured, not asserted: the adjunction round-trip defect of §6 is exactly 0 when the pairwise alignment is a strict adjoint (in every recovered translation and $90°$-rotation case,

$30/30$) and undefined when no adjoint exists (fail-closed, $30/30$, with zero spurious gluings) — a scalar witness for colimit existence.

3. **Decay stays outside the morphisms** (invariant 3), and empirically the trust$\times$staleness gate is unchanged by frame recovery — the same integer breakpoints with and without a shared frame — confirming that the two layers commute as the diagram reading requires.

7.2 Collective memory as a coherent gluing of local place categories

The colimit merge of §7.1 *assumed* the alignment maps and §6 *constructs* them; we can now state, as the central categorical result, what object the collective memory is when agents hold *different* frames, and exactly when it exists faithfully.

Setup. Agent i's realized memory is a thin category (a poset): its concept lattice $L_i = \mathrm{Concepts}(\mathcal{K}_i)$ over i's private frame, objects the places, order the extent inclusion of §7. Frame recovery (§6) yields, on the co-observed subposet, a monotone map $F_{ij} : L_i \to L_j$ that transports i's places into j's frame, and its reverse F_{ji}. Because the categories are *thin*, a recovered pair (F_{ij}, F_{ji}) is an *adjoint pair* precisely when it is a Galois connection, with unit $\eta^{ij} : \mathrm{Id}_{L_i} \Rightarrow F_{ji} \circ F_{ij}$ — the adjunction defect measured in §6.

Definition (coherent alignment system). The family $\{F_{ij}\}$ over the communication graph is *coherent* if (i) each pair is an adjoint order-embedding (η^{ij} an isomorphism, i.e. defect 0: no place is aliased onto a distinct one), and (ii) it satisfies the cocycle condition $F_{jk} \circ F_{ij} = F_{ik}$ on triple overlaps.

Proposition (two agents). For an adjoint pair $F \dashv G$ between L_A, L_B, the collage (comma) category $\mathrm{Gl}(F)$ — objects $L_A \sqcup L_B$, cross-hom $\mathrm{Gl}(F)(X, Y) = L_B(FX, Y) \cong L_A(X, GY)$ (Mac Lane, 1998) — contains L_A, L_B as full subposets and is the two-agent collective memory; the defect η_X is the obstruction to X round-tripping to itself, and the gluing collapses distinct places iff η is not an isomorphism.

Theorem (collective memory = colimit of the alignment diagram). Let $\mathbf{D}$ be the diagram in $\mathbf{Pos}$ (equivalently the enriched $\mathbf{Ctx}_\Sigma$) whose vertices are the L_i and whose edges are the F_{ij}. The collective memory is $\mathrm{colim}\,\mathbf{D}$. As $\mathbf{Pos}$ is cocomplete this colimit always exists; it is a *faithful* gluing — every L_i embeds and no two distinct places are identified — *iff* the alignment system is coherent. When all $F_{ij} = \mathrm{id}$ (a shared frame) $\mathbf{D}$ is the constant diagram and $\mathrm{colim}\,\mathbf{D}$ is exactly the union-find colimit merge of §7.1; the general statement removes the shared-frame assumption by replacing identities with the recovered adjoint embeddings.

Argument. Cocompleteness of $\mathbf{Pos}$ realizes the colimit as the disjoint union quotiented by the preorder generated by the F_{ij}-identifications. Each canonical injection $L_i \hookrightarrow \mathrm{colim}\,\mathbf{D}$ is faithful iff those identifications add no collapse inside an L_i, which holds iff every F_{ij} is an order-embedding (unit iso, defect 0) and the identifications agree across overlaps (cocycle) — i.e. coherence. The shared-frame case is the constant diagram, whose colimit is the componentwise union, i.e. union-find. $\square$

Corollary (two computable witnesses). Coherence is checkable without solving general semantic alignment. (a) The pairwise *adjunction defect* η (§6) is zero iff each edge is a strict adjoint embedding — measured exactly 0 in $30/30$ recovered translation and rotation cases — and guards against *over-merge*. (b) The *connectedness* of the trust$\times$cadence reachability graph (§7.3, Figure 4) decides whether a single $\mathrm{colim}\,\mathbf{D}$ spans all N or the diagram disconnects into a coproduct of sub-consensuses (*fragmentation*). The two failure modes are categorically distinct and separately measured: defect > 0 is collapse, disconnection is fragmentation — the two axes of Figure 4 and the fail-closed behaviour of §6.

Scope. The proofs are the standard cocompleteness of $\mathbf{Pos}$ and the collage construction (Mac Lane, 1998); the theorem's job is to name the object and pin its existence to two measured quantities. The empirical content — that fingerprint recovery actually yields the adjoint embeddings — lives in §6, not in the theorem. One assumption is *not* discharged: a shared tag alphabet Σ; with different

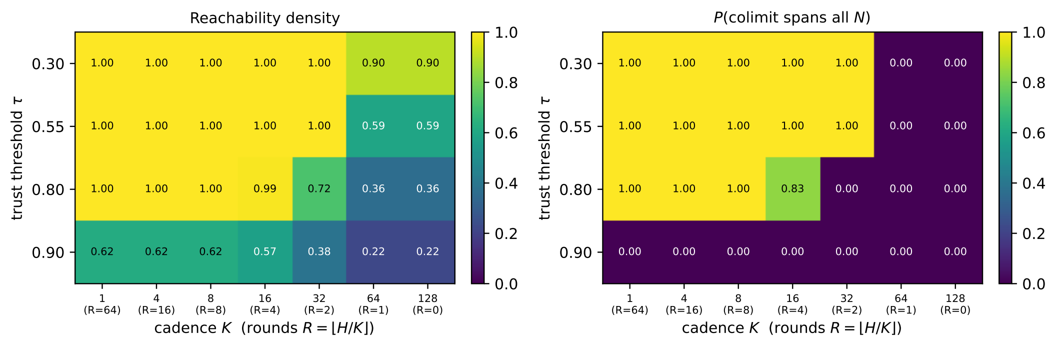

Figure 4: Communication-structure phase diagram over trust threshold τ (rows) and cadence K (columns; $R = \lfloor H/K \rfloor$ broadcast rounds), N=8 agents on an interaction ring, 12 seeds. **Left:** reachability density (fraction of ordered agent pairs whose evidence can compose within R rounds). **Right:** P(the colimit spans all N), i.e. all-to-all reachability. The diagonal boundary is a structural transition: the collective admits a single consensus object in the low-τ/low-K region and fragments in the high-τ/high-K corner, with higher τ demanding lower K (more rounds) to stay connected.

alphabets the F_{ij} would have to translate Σ as well, which is beyond the present recovery and is the honest next frontier.

7.3 Protocol parameters as categorical structure: a trust×cadence phase diagram

The reading so far models the *memory* (objects are semantic contexts, merge is their colimit) under a shared frame and full exchange. Its predictive use comes from turning the lens on the *communication structure* and asking *what shape the colimit takes* as a function of the two protocol parameters — so that the category computes a structural fact rather than re-describing an algorithm. (In $\mathbf{Pos}$ the colimit of the diagram always *exists* by cocompleteness, disconnected or not; the structural question is whether it is a single consensus component or a coproduct of sub-consensuses, exactly the dichotomy of §7.2.) Two parameters play two structural roles:

Trust τ gates morphisms. An edge $i \to j$ (agent i incorporates j's evidence) exists only where i's trust in j is at least τ. Raising τ prunes the trust graph.

Cadence K gates composition depth. Over a horizon H there are $R = \lfloor H/K \rfloor$ broadcast rounds, so a composed chain $i \to k \to \cdots \to j$ can be at most R morphisms long. Low K (many rounds) allows deep composition; high K allows only short paths.

A *consensus-spanning* colimit — a single connected consensus component that fuses and serves every agent's evidence — obtains iff the R-round reachability of the τ-thinned trust graph is all-to-all; otherwise the colimit degenerates into a coproduct of sub-consensuses.

Figure 4 sweeps $\tau \times K$ for N=8 agents whose pairwise trust is produced by the system's own EMA rule (`peer_memory.peer_trust.AsymmetricTrust`) under a ring co-observation topology (neighbours co-observe often and earn high trust; distant agents do not), averaged over 12 seeds (`experiments/collective_semantic_memory/phase_diagram_trust_cadence.py`). The result is a genuine phase transition with a diagonal boundary. At low τ the trust graph is dense and the colimit spans all agents at every cadence. Raising τ prunes the long-range morphisms down to a near-neighbour ring; the collective then keeps a spanning colimit *only if* cadence is low enough for multi-hop composition through still-trusted intermediaries to reconnect it — τ=0.55 holds the spanning colimit down to R=2, τ=0.80 needs $R \geq 4$ (the ring diameter), and τ=0.90 fragments throughout. Cadence and trust are therefore not opaque knobs: together they determine a categorical structural invariant, and the phase boundary is exactly where the collective transitions from a single consensus object to fragmented sub-consensuses.

This is the sense in which the categorical model is operational rather than descriptive: each protocol parameter maps to a concrete change in the communication diagram, and the diagram's structural invariant (whether the colimit spans all agents or splits into a coproduct) predicts a qualitative regime of the collective. *Scope.* The trust matrix is produced by the real EMA rule under a modelled ring co-observation topology, and “consensus-spanning” is operationalised as all-to-all R-round reachability; establishing quantitatively that crossing this structural boundary predicts the drop in observed Q/R/M/C performance is the natural next experiment.

7.4 Toward symbol alignment: grounding tag alphabets through shared places

The gluing theorem (§7.2) still assumes a shared alphabet Σ. Dropping it, each agent holds a context (G_i, Σ_i, I_i) over its *own* vocabulary, and the right morphism is no longer a functor over one Σ but an *infomorphism* (Barwise and Seligman, 1997): a pair $f : G_A \to G_B$, $\hat{f} : \Sigma_B \to \Sigma_A$ with $I_B(fg, y) \Leftrightarrow I_A(g, \hat{f}y)$. Infomorphisms compose and have colimits (Barwise–Seligman channels), so the collective memory over heterogeneous alphabets is the *channel core* — the colimit of the infomorphism diagram — which extends §7.2 verbatim (functors over one Σ become infomorphisms over the Σ_i; shared Σ with $\hat{f} = \mathrm{id}$ is the old case).

Grounding mechanism. Symbols align through shared referents: on the places already anchored by frame recovery (§6), the co-occurrence of an A-tag and a B-tag is evidence they name the same feature, and $\hat{f}$ is read off the anchored co-occurrence. (When the alphabets differ so much that fingerprints themselves cannot seed the anchors, one bootstraps from the alphabet-independent structural constellations and alternates recovery of f and $\hat{f}$.) The *symbol-alignment defect* def_Σ — the fraction of anchored (place, y) pairs violating the infomorphism biconditional — is the symbol analogue of the frame adjunction defect of §6.

Preliminary result. A controlled study (`experiments/warp/symbol_alignment.py`; $|\Sigma_A|=8$, 120 places, 20 seeds) gives agent B a secret relabelling of A’s alphabet with observation noise, and recovers $\hat{f}$ from anchored co-occurrence.

Table 2: Symbol alignment by grounding vs. assuming identical alphabets. Target-ID F1 is identifying A’s target feature from B’s (translated) evidence. “naive” assumes the alphabets coincide by index; “recovered” learns $\hat{f}$; “oracle” is the true map.

noise	transl. acc.	def_Σ	F1 naive	F1 recovered	F1 oracle
0.00	1.00	0.000	0.56	1.00	1.00
0.10	1.00	0.099	0.55	0.92	0.92
0.30	0.89	0.304	0.56	0.69	0.73

Assuming identical alphabets fails (F1 ≈ 0.55: the relabelling sends B’s target tag to the wrong feature); grounding recovers the map (translation accuracy 1.00, def_Σ tracking the noise floor) and reaches oracle-level target identification, robustly down to 15% anchored places (accuracy 0.95). Coarsening two features into one B-tag bounds the achievable accuracy — there is no faithful bijection, so the channel core coarsens rather than errs. Symbol alignment is thus not a wholly open problem but the same grounding-plus-defect recipe one alphabet up; a full treatment (genuine incommensurability, online drift) is the next paper.

8 Limitations and next experiments

Grid substrate. The current result is intentionally controlled: all experiments are in grid worlds. A continuous-state substrate such as VMAS, Habitat, ProcTHOR, or MiniWorld is the next portability step.

Coordinate anchor. Removed in Section 6: local fingerprint signatures recover the inter-agent frame (translation and $90°$ rotations) exactly, and the merge/trust/staleness results are unchanged on top of the recovered frames. What remains open is the harder end of the same axis: continuous $SE(2)$, appearance noise (the current matcher uses exact constellation equality because the substrate is noise-free). Symbol alignment — agents with *different tag alphabets* — is addressed preliminarily in §7.4 (grounding through shared places, recovered as an infomorphism); genuine incommensurability and vocabulary drift remain open.

Minimal CSM. The CSM rules are deliberately simple. Adaptive cadence K_t and intent-conditioned trust are the next architecture variants. Selective consolidation of under-represented tags now has a measured pilot: in the companion route-transfer paper, a purely relational *song* (landmark signatures + beat vectors, no global coordinates) carries the same never-seen-target transport as a full snapshot at 6.4% of the bit cost, fail-safe — beats alone fail open, signatures alone cannot reach the unseen. Replacing snapshot exchange with budgeted songs, merged under the same three CSM rules, is therefore no longer speculative; making the song grammar (what triggers a couplet, what enters a signature) subject to selection rather than design is the open half.

Learned baselines. CommNet-style learned communication is a useful comparison, but the more relevant future baseline is an explicit memory LLM or vector-memory multi-agent system measured on the same task.

9 Conclusion

Collective memory is not monotone. In scarce multi-agent navigation, instant information sharing can reduce completion by causing agents to converge on the same targets. Peer cadence exposes the trade-off, but cadence alone is a one-dimensional control surface. A collective semantic memory should make merge, trust, and staleness explicit. The minimal CSM does exactly that and moves the system to a new operating point: neither fast-broadcast M-saturation nor slow-broadcast C-saturation, but a more balanced M$\times$C profile with higher success than every fixed peer cadence in the scarcity benchmark.

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A The coalgebraic layer: individuals as codata, the collective as a final coalgebra

The reading of §7 fixes belief *content*: at a state, the concept lattice $L_t^i = \mathrm{Concepts}(\mathcal{K}_t^i)$ says what is retrievable, and the merge of §7.1 says how peer contents combine into $\widetilde{\mathcal{K}}_t^i$. Both are *algebraic*: they build a belief object. Neither speaks to the agent as a non-terminating *process* that must emit an action and a signal at every tick under a moving environment, nor to the sense in which a *population* comes to share a convention. That process side is *coalgebraic* (Rutten, 2000; Jacobs, 2016). We make it precise on the water-search example and show the two sides are dual, meeting inside a single state transition. (The belief/desire/intention vocabulary follows the BDI agent tradition (Rao and Georgeff, 1995), here given coalgebraic content.)

Signature. Fix an action set Act (move, dig, noop), a per-agent observation set O_i (agent i's local percept — its *unique view*), and an *ostension* set

$$M \;=\; 1 \,+\, (X \times \Sigma),$$

whose values are either *silence* (the summand 1) or a pointer "place x carries tag σ" (whistle-and-point at a water-marked cell). That silence is a first-class value of M matters: the ostension channel *can be informatively empty*, and it is separate from the state update below. This is exactly the observability that a learned real-valued message vector destroys (its channel is never empty and never separates signalling from stepping), and it is why the symbolic stages remain readable.

State. We take the coalgebra carrier to be the H-MDP high-level state of §7,

$$\mathcal{S}_i \;=\; \big\{\, h = (\mathcal{K}^i,\, D^i,\, g^{i,\star},\, s^i)\,\big\},$$

carrying the same information as $(L^i, D^i, g^{i,\star}, s^i)$ since $L^i = \mathrm{Concepts}(\mathcal{K}^i)$. Thus *belief lives in the state*; the lattice is a projection of it.

Definition (codata: the agent's cotype). The observable behaviour of agent i is a coalgebra for the endofunctor

$$B_i(S) \;=\; \underbrace{\mathrm{Act}}_{\text{act now}} \;\times\; \underbrace{M}_{\text{signal now}} \;\times\; \underbrace{S^{\,O_i \times M^{A\setminus\{i\}}}}_{\text{continue given (my next percept, peers' signals)}},$$

i.e. a map $c_i \;=\; \big\langle\, \texttt{observe\_act}_i,\; \texttt{observe\_ostension}_i,\; \texttt{step\_intension}_i \,\big\rangle \;:\; \mathcal{S}_i \;\to\; B_i(\mathcal{S}_i)$ with three destructors

$$\texttt{observe\_act}_i : \mathcal{S}_i \to \mathrm{Act}, \quad \texttt{observe\_ostension}_i : \mathcal{S}_i \to M, \quad \texttt{step\_intension}_i : \mathcal{S}_i \to \mathcal{S}_i^{O_i \times M^{A\setminus\{i\}}}.$$

Here $M^{A\setminus\{i\}} \;=\; \prod_{j\neq i} M$ is the tuple of peers' ostensions delivered this tick. Operationally, $\texttt{step\_intension}_i(h)(o, (m_j)_{j\neq i})$ (i) folds the percept o and the peer pointers (m_j) into the context by the trust/staleness merge, $\widetilde{\mathcal{K}}_t^i = \mathrm{Merge}_{\mathrm{trust,stale}}(\cdots)$ of §7.1; (ii) recomputes the lattice; (iii) re-evaluates the desire predicate D^i; and (iv) re-locks the intention $g^{i,\star} = \chi_t^i(\mathrm{ext}(D^i))$ of §7. So Merge and χ are *components of one destructor*. The agent *is* the codata generated by c_i: it has no finite terminal plan, only the standing ability to emit an action and an ostension and to continue. This is the "infinite survival generator" of the motivating example.

Definition (indexed family). The system is the A-indexed family $\{(\mathcal{S}_i, c_i)\}_{i\in A}$, a coalgebra in $\mathbf{Set}^A$. The functor B_i depends on the index i through O_i (and through the peer set $A\setminus\{i\}$): the same survival cotype is instantiated at each agent through that agent's local viewpoint. Agent 1 on the dune generates from "I see the horizon"; agent 2 in the hollow generates from "I feel the soil." The index is the viewpoint.

Definition (codependence: the collective coalgebra). Let the environment deliver a joint percept $o = (o_i)_{i\in A} \in \prod_i O_i$. The collective one-tick map on $\prod_i \mathcal{S}_i$ wires each agent's ostension output into the others' step inputs,

$$\Phi\big((h_i)_i\big)(o) \;=\; \Big(\, \texttt{step\_intension}_i(h_i)\big(o_i,\, (\texttt{observe\_ostension}_j(h_j))_{j\neq i}\big) \,\Big)_{i\in A}.$$

Because i's next state is a function of the *destructor outputs* of the others, the family is not N independent processes but one coalgebra for the product functor $\mathbf{B}(S) = \big(\prod_i(\mathrm{Act}\times M)\big)\times S^{\prod_i O_i}$ on $\prod_i \mathcal{S}_i$. This wiring is the *codependence*: agent 1 cannot step without passing agent 2's ostension through its own `step_intension`. Broadcast cadence enters here as a gate: at non-broadcast ticks the delivered tuple (m_j) is the last received (stale) tuple, or silence; K is the delivery period.

Definition (behaviour and bisimulation). B_i is a container (polynomial) functor, so it has a final coalgebra $(\nu B_i,\ \mathrm{beh}_i)$ (Aczel, 1988; Rutten, 2000) whose elements are the $\big(O_i\times M^{A\setminus\{i\}}\big)$-branching, $(\mathrm{Act}\times M)$-labelled behaviour trees, and there is a unique coalgebra morphism $!_i : \mathcal{S}_i \to \nu B_i$ sending a state to its whole future behaviour. A B_i-*bisimulation* is a relation $R \subseteq \mathcal{S}_i \times \mathcal{S}_i$ under which related states emit equal action and equal ostension and, on every input, step to R-related states. Container functors preserve weak pullbacks, so the *greatest bisimulation* $\sim_i$ exists, is an equivalence, and equals $\ker(!_i)$: two states are bisimilar iff they have the same behaviour (Sangiorgi, 2011). The same holds for the collective coalgebra $(\prod_i \mathcal{S}_i,\ \langle(\mathtt{observe\_act}_i)_i, (\mathtt{observe\_ostension}_i)_i, \Phi\rangle)$, with greatest bisimulation $\approx$.

Definition (society / culture). The *collective* is the image of the states in the final coalgebra $\nu\mathbf{B}$, equivalently the quotient $\big(\prod_i \mathcal{S}_i\big)/\approx$. It is neither a central server nor a shared map: it is a bisimilarity class of joint behaviour. A *water-search convention* is this invariant read on the water sub-signature. Let $M_{\mathrm{water}} \subseteq M$ be the pointers whose tag lies in the water family, and let $\nu B_i \to \nu B^{\mathrm{w}}$ be the forgetful map to the behaviour trees restricted to receiving M_{water} and emitting the corresponding intention lock. The population *holds the convention* when all agents' reachable behaviours agree under this map: hearing "point-and-dig at a water-marked place" drives every agent's intention destructor to lock that place (up to occupancy). Governance is then automatic in both directions. A new agent (the example's fourth agent) is a member of the society exactly when its behaviour lands in the class $[\cdot]_\approx$ on M_{water} — otherwise its ostensions are not understood and its own step never converges (top-down); and the class itself was reached by the local, coupled steps of the first agents at one green shrub (bottom-up).

Remark (communication = lazy, depth-one corecursion). The behaviour $!_i(h)$ is defined by unfolding c_i indefinitely. On receiving $m_j = \mathtt{observe\_ostension}_j(h_j)$, agent i uses only this *depth-one* destructor output inside $\mathtt{step\_intension}_i$; it does *not* compute $!_j(h_j) \in \nu B_j$, the peer's full behaviour, which is the "I think that you think…" regress. Communication is therefore corecursion truncated to the least depth that resolves the receiver's own next step — a finite approximation of the final-coalgebra map — with a relevance-theoretic stopping rule (Sperber and Wilson, 1995) deciding that depth. The regress collapses into one effective action.

Remark (algebra–coalgebra bridge to §7–7.1). Belief is algebraic: the FCA lattice L^i_t built by the colimit merge is the *content at a state*. The agent is coalgebraic: the codata observed by the three destructors is the *process over states*. They are dual and meet in one place — the colimit merge and the satisficing choice χ are the internals of `step_intension`. The memory-growth diagram $\mathcal{K}^i_0 \to \mathcal{K}^i_1 \to \cdots$ of §7 is precisely the belief-projection of the coalgebra trajectory $h^i_0 \mapsto h^i_1 \mapsto \cdots$.

Coalgebraic reading of the bottleneck shift. The result of §4 and Table 1 has a one-line reading here. Two kinds of bisimilarity compete. Bisimilarity on the *ostension / belief* sub-signature is the wanted invariant: it is the shared water convention. Bisimilarity on the raw *intention* output over a *scarce* target is the unwanted one: several states mapping to the same $g^\star$ is a collision. Occupancy $\mathcal{O}_{-i}$ is the extra-coalgebraic constraint that penalises the second. Cadence K tunes how hard the coupling Φ drives the collective toward the *full* greatest bisimulation $\approx$: small K delivers fresh ostensions often, enlarging the merged water extent so intention locks reliably ($P(M^\star \mid R^\star)\uparrow$), but over-synchronises the intention destructor so agents collide ($P(C^\star \mid M^\star)\downarrow$). The interior optimum $K{=}8$ is where belief converges without intention collapsing onto one cell; the minimal CSM adds trust and staleness precisely so that `step_intension` reaches belief-bisimilarity (fresh, trusted, aligned evidence) while damping intention-bisimilarity (stale evidence decays before it can herd every agent onto the same lock). We state this as a reading consistent with the data, not as a theorem: the container functors, their final coalgebras, and bisimulation are standard constructions used here as a lens, and the relevance-theoretic stopping rule is an interpretive gloss rather than a fitted mechanism.

Table 3: The motivating example's terms as formal objects of the coalgebraic reading. Each row is the water-search meaning of a construction already used in §A.

Informal term	Formal object	Water-search meaning
Cotype (codata)	B_i-coalgebra $(\mathcal{S}_i, c_i)$	the agent's non-terminating action/signal generator
Indexed type	family over A; dependence of B_i on O_i	the agent's unique local viewpoint (dune vs. hollow)
Codependent type	wiring $\texttt{observe\_ostension}_j \to \texttt{step\_intension}_i$ in Φ	"I step only through your ostension"
Individual (instance)	a state $h \in \mathcal{S}_i$ and its behaviour $!_i(h)$	the running agent digging and whistling now
Destructors (cogenerators)	`observe_act`, `observe_ostension`, `step_intension`	step-and-dig; whistle-and-point; fold neighbour's signal into next state
Lazy evaluation	depth-one corecursion + relevance stop	unfold the peer's coalgebra one step, not its whole behaviour
Final coinductive core / society	$\nu\mathbf{B}$; greatest bisimulation $\approx$ (on M_{water})	the stable, distributed water-search culture

B Context is not memory: raw context vs. the Songlines graph under one model

Modern LLMs *accept* long context; the question this appendix measures is whether they turn it into a stable, globally coherent state — and whether the Songlines graph supplies that state instead. In every arm the *same* model answers; only the memory substrate differs: **raw** (the full observation transcript in the prompt — typical usage) vs. **graph** (the same evidence consolidated by the Songlines rule: latest state per place, staleness explicit, $O(\text{salient places})$ prompt). Three axes (`experiments/context_vs_structure/`; HF backend on SLURM GPU nodes and local Ollama as an independent second backend).

Axis 1: volume and integration (one witness log). A witness log over a 30×24 grid contains a freshness conflict (water seen repeatedly at site A, later dried; one fresher sighting at B). Questions: *fact* (where is water now), *count* (how many waters now), *hop* (nearest hazard to current water).

Table 4: Accuracy by question, arm, and log volume L (lines). Qwen2.5-7B (cluster, 20 seeds/cell) on cells within its compute envelope; Llama-3.1-8B (local, 10 seeds/cell) at large L with *verified full ingestion* (`prompt_eval_count`=25,030 tokens at L=2000; no truncation). Graph prompts stay ≈350 bytes at every L.

Question	Arm	L=50	L=200	L=800	L=2000
fact	raw	1.00 / –	1.00 / –	– / 0.00	– / 0.00
fact	graph	1.00 / –	1.00 / –	1.00 / 1.00	1.00 / 1.00
count	raw	**0.00** / –	**0.00** / –	– / **0.00**	– / **0.00**
count	graph	1.00 / –	1.00 / –	1.00 / 1.00	1.00 / 1.00
hop	raw	0.00 / –	0.05 / –	– / 0.00	– / 0.00
hop	graph	0.00 / –	0.15 / –	0.40 / 0.50	0.30 / 0.60

Two failure layers, both with identified mechanisms. *Cognitive:* the raw arm loses global state at **any** volume — on *count* it answers "2" (17/20) or "3" (3/20) when the truth is 1: it adds up historical mentions instead of resolving freshness, i.e. no consolidation. At large L even single-fact retrieval collapses despite verified full ingestion (a probe with the answer at the very end of a 25k-token prompt is also missed). *Systemic:* on the cluster's 24 GB sm75 GPUs, raw-arm inference at 5–10k tokens exhausts memory through the quadratic attention matrix (marked as compute-envelope cells, "–", never reported as cognition), while the graph arm runs at constant cost at every volume. Raw context loses twice — as cognition and as systems cost; *hop* shows the honest boundary: structure repairs retrieval, not arithmetic reasoning.

Axis 2: time (sessions). A persistent world drifts (water relocates, p=0.35/session); after each of 8 sessions the model is asked where the water is now, from **raw replay** (all past transcripts re-injected, the typical chat pattern) vs. the persisted **graph**. Qwen2.5-7B: the graph arm holds 0.62–1.00 across all eight sessions at a flat $\approx$0.3 kB prompt; the replay arm's valid cells are already at 0.00–0.67 in the earliest sessions and its prompt grows to 25 kB by session 8 (later cells envelope-limited on the cluster; on the local backend, where no envelope exists, replay degrades to 0.00–0.50 by 19 kB — the same pattern).

Axis 3: system size (multi-agent, private frames). Three agents accumulate observations over G sessions, each logging in its *own* secret coordinate frame. Question: "how many DISTINCT water sources exist?" — the same physical water seen by two agents at different private coordinates must be counted once. The raw arm gets all three logs (and is told the frames differ); the graph arm gets the Songlines machinery's merged map (fingerprint frame recovery, §6, plus dedup). Qwen2.5-7B: graph accuracy *rises* with growth — $0.40, 0.90, 1.00, 1.00$ at G=$1, 2, 4, 8$ ($\approx$330 bytes) — because frame alignment locks as fingerprints accumulate (the exact% curve of the growth study); the raw arm (13k–105k chars) does not produce a single correct answer on any envelope-valid cell. Cross-frame identity is precisely what a context window cannot represent and the gluing structure can.

Summary. The same model, given the same evidence, preserves long, drifting, multi-agent context when — and only when — the evidence reaches it as the consolidated Songlines structure: flat-cost prompts, freshness resolved, frames glued. Given the raw stream instead, it fails on integration at 1.3 kB, on freshness at 19 kB with full ingestion, and on cross-frame identity at every tested size. Scope: two model families, one task family per axis, and the compute-envelope cells are reported as such; the graph arm's consolidation here is the deterministic Songlines rule — LLM-driven consolidation is the roadmap's next stage.

C Reproducibility commands

- Main cadence sweep: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- Extended cadences: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`
- Q/R/M/C analysis: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- Minimal CSM benchmark: `python-mexperiments.collective_semantic_memory.run_csm_benchmark--out_dirtmp/csm_benchmark`
- MiniGrid portability sweep: `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`
- Adjunction round-trip defect of frame recovery (Section 6): `pythonexperiments/warp/alignment_defect.py`
- Context-vs-structure suite (Appendix B): `experiments/context_vs_structure/run_ctx_vs_struct.py`, `run_persistence_llm.py`, `run_growth_llm.py` (HF backend on GPU or –backend ollama locally)
- Trust×cadence structural phase diagram (Section 7.3, Figure 4): `pythonexperiments/collective_semantic_memory/phase_diagram_trust_cadence.py--figfigures/fig_phase_diagram.pdf`

D Asset and license note

The experiments use MiniGrid and MiniWorld (Chevalier-Boisvert et al., 2023). Locally implemented benchmark and memory code will be released under the MIT License as part of the artifact package.

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